

Computer Science & Networking MSQ Practice Exam

Official Study Material & Exam Questions

1. Which of the following are operating systems?

- A. Windows
- B. Linux
- C. Chrome
- D. Ubuntu

Answer: A, B, D

Explanation: Windows, Linux, and Ubuntu are operating systems. Google Chrome is a web browser.

2. Which of the following are relational database management systems?

- A. PostgreSQL
- B. MySQL
- C. Redis
- D. Oracle Database

Answer: A, B, D

Explanation: PostgreSQL, MySQL, and Oracle are relational (SQL) databases. Redis is an in-memory key-value store.

3. Which of the following are application layer protocols in the OSI / TCP-IP model?

- A. HTTP
- B. FTP
- C. TCP
- D. DNS

Answer: A, B, D

Explanation: HTTP, FTP, and DNS operate at the Application Layer (Layer 7). TCP operates at the Transport Layer (Layer 4).

4. Which of the following are network security devices or systems?

- A. Firewall
- B. IDS (Intrusion Detection System)
- C. IPS (Intrusion Prevention System)
- D. Compiler

Answer: A, B, C

Explanation: Firewalls, IDS, and IPS are security systems designed to protect networks. A compiler translates programming language source code into machine code.

5. Which of the following data structures have $O(1)$ average time complexity for insertion?

- A. Hash Table
- B. Singly Linked List (at head)
- C. Binary Search Tree (unbalanced worst case)
- D. Array List (insertion at arbitrary middle index)

Answer: A, B

Explanation: Hash tables average $O(1)$ insertions, and inserting at the head of a linked list is $O(1)$. BST insertions take $O(\log n)$ average time.

6. Which of the following are symmetric encryption algorithms?

- A. AES (Advanced Encryption Standard)
- B. DES (Data Encryption Standard)
- C. RSA
- D. Blowfish

Answer: A, B, D

Explanation: AES, DES, and Blowfish use symmetric keys. RSA is an asymmetric (public-key) cryptosystem.

7. Which of the following are recognized cloud computing service models?

- A. IaaS (Infrastructure as a Service)

- B. PaaS (Platform as a Service)
- C. SaaS (Software as a Service)
- D. CaaS (Compiler as a Service)

Answer: A, B, C

Explanation: The NIST cloud model defines IaaS, PaaS, and SaaS as the primary service delivery models.